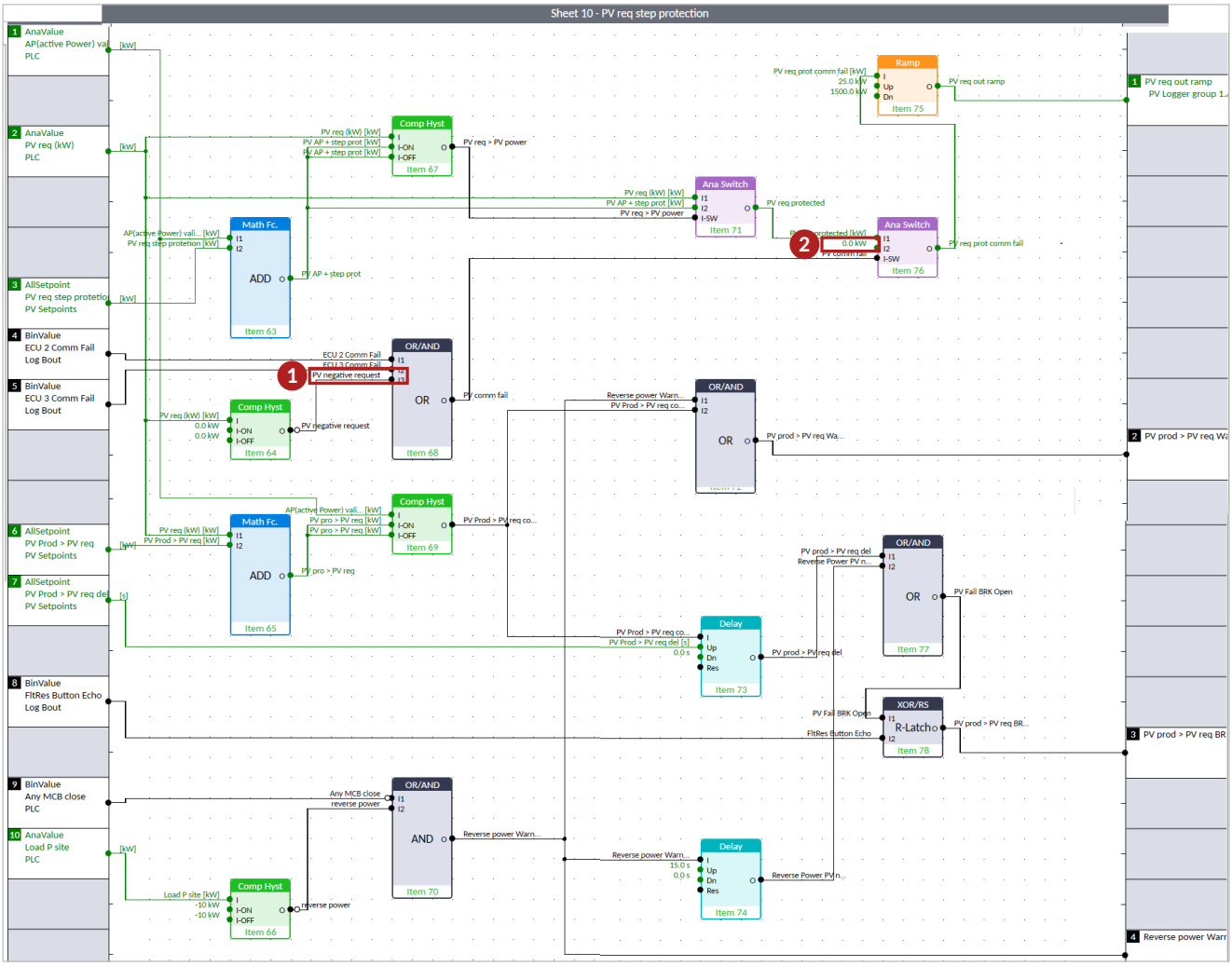
Sheet 10 today, and what goes wrong

Erbil plant - InteliNeo 5500 master - PLC sheet 10 "PV req step protection"



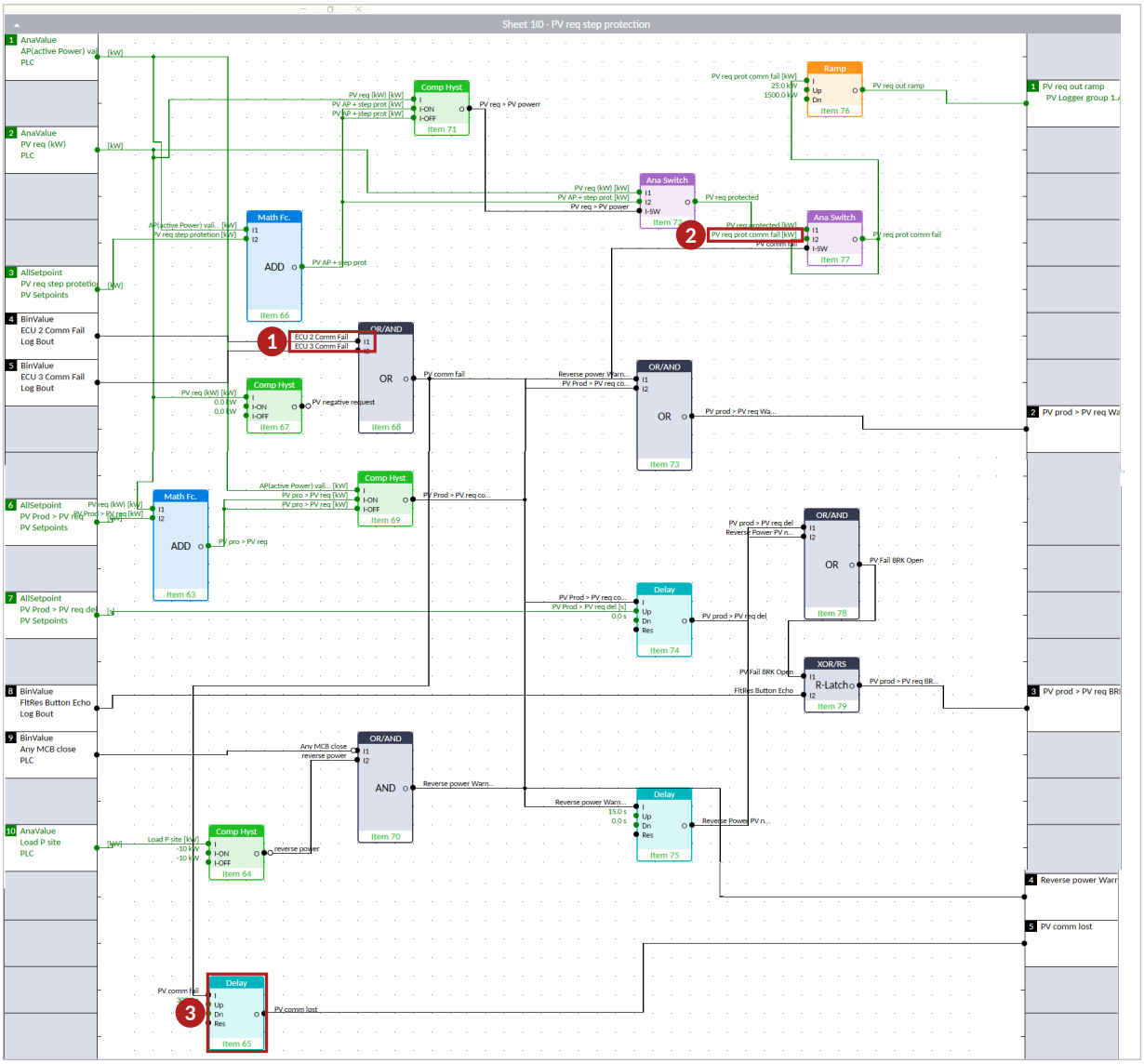
This sheet sends the solar plant its setpoint. Most of it is fine. The problem is one small switch, Item 76. Normally it just passes the request through. But if the controller misses even one reply from the PV logger, Item 68 (1) flips the switch, and the other side of the switch is a fixed 0.0 kW (2). So the solar plant gets told: zero.

The inverters do what they are told, within seconds. The gensets were running at 25 % and now they carry everything, in one step. Then the reply comes back, and the solar has to climb up again, 100 kW at a time, about five minutes. On 14 October, my last check, the history shows it 20 times between 08:52 and 09:58.

While the link is down, the zero never even reaches the logger. It arrives when the link is back, so it protects nothing. We just lose production every time.

What I want to change

Erbil plant - InteliNeo 5500 master - PLC sheet 10 "PV req step protection"



Three small edits, nothing else. The switch (now Item 77) gets its own output on its second input (2). So when a reply is missed, it just keeps sending the last setpoint. Nothing drops, nobody notices. Item 68 (1) now only listens to the two real comm fail signals, so if the controller really wants zero, zero still goes out. And one new timer, Item 65 (3), raises a flag called PV comm lost if the logger stays silent for five minutes. For now that flag only goes to an alarm.

Why I think we should do it

With a healthy link, the program does exactly what it does today. I ran the whole program block by block against our own history to check that. While the link is down, nothing is different from today either: the inverters keep their last setpoint in both versions, the gensets have 25 % in hand, the controller's own protections still work without the logger, and after five minutes we get the flag. If anyone wants it back, we load the archive from 14 October. One minute.

What it is worth

I ran 14 October through the model twice, with and without the change. With the plant run the way it is today (three gensets by hand) the change gives about 140 kWh more solar and saves about 45 litres of diesel on that day. With the controller's power management switched on, the same change is worth about 350 kWh and 100 litres a day. The real saving is the next step, running fewer gensets, roughly 1,300 litres a day; this change is what makes that step safe.